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**PROJECT TASK 3: CHOOSING THE APPROPRIATE LAN
DEVICES**

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GROUP 5

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Table of Contents

3.1 Introduction to the Needed Devices	3
3.2 Table of Devices	8
3.3 Reflections	9
3.3.1 Are you surprised by the prices? How were you surprised?	9
3.3.2 Have you ever considered cost as a factor for choosing networking devices?	9
3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.	9
Meeting Minutes #1	21

3.1 Introduction to the Needed Devices

1. Router

Routers can be regarded as the center of connection of different networks from local networks, such as those in buildings merged with the interconnectivity of the internet. They provide communication across different networks, service data transmission from the sender to the receiver nodes, and ensure packet delivery. Enterprise-grade routers have been installed to provide reliability and support dynamic routing protocols and scalability.

2. Switch

Switches form the LAN's backbone within the building by connecting all end-user devices such as PCs, projectors, smartboards, and IoT devices. In each lab, managed switches with VLAN capabilities will be installed to finalize traffic separation by department or function, thus decreasing congestion to boost efficiency.

3. Patch Panel

Patch panels will be used to centralize and organize network cabling within the server room and labs. The patch panels will facilitate cable management and enable easy reconfiguration and troubleshooting by providing a structured connection point for network cables. Each patch panel will handle connections for wired devices in the labs, including PCs, projectors, and servers.

4. Cable

The cabling represents the foundation of the physical infrastructure of the network inside the building. The project will use Fiber-optical cables for the network backbone to ensure high-speed data transfer across different lab infrastructures with faster protocols and least latencies. This is to support a very high-bandwidth application such as network simulation and video streaming. Besides, Cat6 Ethernet cables for workstation connections, supporting speeds up to 10 Gbps within short distances. These will be used for desktop PCs and other peripherals in labs and offices.

5. Modem

The modem connects the internal network of a building to the Internet Service Provider. It changes a digital signal to analog for it to pass through the ISP's infrastructure and vice versa. A high-performance modem is critical for maintaining consistent internet access for bandwidth-intensive activities such as online learning, virtual meetings, and collaborative projects.

6. Access Point

Wireless connectivity is fundamental to modern academic environments, especially in student lounges, where students bring personal devices such as laptops and tablets. The project will implement Wi-Fi 6 access points, providing faster data rates, lower latency, and higher channel efficiency in high-density traffic environments. These access points will allow seamless connectivity on campus, with full coverage of labs, classrooms, and common areas.

7. Server

Servers are essentially a backbone of centralized operations and provide supporting applications, managing data storage and networking applications. In this project, a dedicated server in the Cisco Network Lab will allow students to gain hands-on experience with network simulation tools. The room will house additional servers that will be required for the Embedded Lab and hybrid classroom data management. The intention is to make optimal use of the server resources thereby ensuring data management security.

8. Firewall

Secured networks have always been extremely important especially due to the ever-increasing reliance on electronic media. A hardware-based firewall will be deployed to safeguard the network from external and internal threats. It will monitor and filter traffic between the internet and the internal network, protecting sensitive academic and operational data from unauthorized access, malware, and other cyber threats.

9. PC

A personal computer, also known as a PC, is a multi-purpose microcomputer designed for individual use, with size, capabilities, and price that make it feasible for personal use. It usually consists of a computer case, power supply, motherboard, storage devices such as hard disk drives or solid-state drives, and often an optical disc drive. External devices commonly used include monitors, keyboards, and mice. PCs are designed to be used directly by the end users such as word processing, browsing the internet, email, play multimedia, and games without requiring computer experts or technicians. In contrast with the large and expensive minicomputers and mainframes, PCs are not designed for multitasking by several users at the same time.

10. Monitor

A display device that shows visual information coming from a PC where the user can interact directly with an application or any content. Most modern monitors boast LCD screens with LED backlighting that replaced the older CCFL backlit displays. They connect to computers via DisplayPort, HDMI, USB-C, DVI, or VGA; others make use of proprietary connectors. A good monitor enhances clarity for text, images, and videos. In graphic design and video editing, high-resolution monitors can provide much higher quality compared to their low-resolution siblings. Adjusting stands, anti-glare screens, and higher resolutions (4K or above) also offer greater comfort and productivity. They enable great multitasking with dual and multiple monitor setups.

11. Projector

A projector basically shows images or videos on bigger screens for mass use when required for presentations, lectures, and discussions. They accommodate larger audiences by sharing all videos or slide representations clearly. Modern projectors give special prominence to 4K display abilities alongside wireless connectivity and multiple means of input device compatibility with state-of-the-art integration. Some projectors work by projecting images directly with lasers, while others shine light through a transparent lens. There are also portable projectors for those presentations that must be on the go. A virtual retinal display projects images directly onto the retina instead of using an external screen.

12. Smart TV

A Smart TV, also known as a connected TV or Hybrid Television, is a television set that comes with integrated internet and interactive features. It can stream music, videos, browse the web, and view photos. Smart TVs integrate traditional TV features with internet-based services, such as streaming and radio. They run on an operating system with applications and digital content, which can be updated through an app store. This technology is also found in set-top boxes, game consoles, and smartphones, which allow access to content from the web, broadcast TV channels, or local storage.

13. Smartboard

A smartboard is a large, usually wall-mounted interactive display board in the form of a whiteboard. It can be either a standalone, used independently, touchscreen computer used to perform tasks and operations or a connectable apparatus that is used as a touchpad to control computers from a projector. They are small, touch-screen-enabled computers. They are applied in a number of environments, from classrooms at all levels of education to corporate boardrooms and work groups, training rooms for professional sports coaching, broadcasting studios, and others.

14. Microphone

A microphone converts sound into an electrical signal for communication, recording, and broadcasting. It's essential for virtual meetings, presentations, and events, ensuring clear sound. There are different types for specific uses: handheld microphones for interviews or performances, lapel microphones for hands-free use, and condenser microphones for sensitive studio recording. Noise-canceling microphones reduce background noise for clearer audio, even in noisy settings. Many microphones also have wireless connectivity and adjustable sensitivity.

15. Connectors

Connectors serve a purpose of connecting different devices like PCs, monitors, projectors, and other peripherals to enable easy data transfer and compatibility between devices. Common connectors include HDMI, USB, VGA, and DisplayPort, while

modern connectors also feature wireless technologies such as Bluetooth and Wi-Fi, reducing cable clutter and allowing more flexible device placement. Adapters and docking stations further expand connectivity options. Optical fiber connectors are used to link optical fibers and allow the efficient transmission of light signals, thus offering quicker connection and disconnection compared to splicing.

16. Fiber Optic Cable

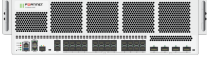
A fiber-optic cable, also called an optical-fiber cable, is an assembly containing one or more optical fibers that are used to carry light signals. The optical fiber elements are usually individually coated with plastic layers and enclosed in a suitable protective tube for the environment of deployment. Fiber optic cables are crucial in high-speed internet connectivity, granting fast and reliable access to applications such as video conferencing, online research, and cloud computing. They exhibit low latency, high bandwidth, and are much more durable and secure than conventional copper cables. There are various types of fiber-optic cables for different applications, which include long-distance telecommunication to high-speed data connections inside buildings. These devices combined facilitate modern setups for education, business, and personal use, creating a cohesive environment that is productive.

3.2 Table of Devices

Device	Description	Price (RM/unit)	Quantity	Total Price(RM)
Router 	Brand: TP-Link Model: Archer AX6000 Features: • Dual-Band Wi-Fi 6 • Quad-core 1.7 GHz processor • MU-MIMO and OFDMA support • Trend Micro malware protection • 3-year limited warranty	1,029.00	14	14,406.00
Switch 	Brand: D-Link Model: DGS-1250-52X Features: • 52 ports (48 Ethernet, 4SFP + 10 Gbps ports) • 176 Gbps switching capacity • Maximum forwarding rate of 130.95 Mpps • PoE support for powering connected devices • 16 Mbits packet buffer memory for efficient data management	2356.00	14	32984.00
Patch Panel 	Brand: FS Model: Cat6 24-Port Patch Panel Features: • Support RJ45, HDMI and USB connections • Compatible with Cat6 and Cat5e cables	312.00	10	3120.00

	• Shielded and unshielded versions to protect against interference • 24 ports for large-scale connectivity • Rack-mounted design for easy installation and organization			
Cable 	Brand: Cable Matters Model: Riser Rated (CMR) Cat6 Bulk Ethernet Cable Features: • Cat6 Bulk cable that need cutting, crimping and termination • 1000 feet (305 meter) • Speed up to 10GBps and bandwidth up to 550MHz • Pure copper conductors that enhanced able performance • Convenient easy-pull reel to ease installation process. • Unshielded Twisted Pair (UTP) • Riser-rated PVC jacket for durability and fire resistance • Consist multiple colors for cable management and identification • Compatible with Ethernet, switches, routers, patch panels, and PoE devices	1888.73 /1000ft (305 meter)	10	18888.73
Modem	Brand: ARRIS Model: Surfboard SB8200 DOCSIS 3.1 Features: • DOCSIS 3.1 standard • Dual Gigabit Ethernet ports • Future-proof design for	755.77	5	3778.85

	compatibility with modern high-speed networks ● Ideal for high-bandwidth activities ● Reliable performance for multiple devices in busy environments			
Access Point 	Brand: TP-Link Model: EAP610 Ultra-Slim Features: ● Dual-band Wi-Fi 6 with speeds up to 1800 Mbps ● MU-MIMO, OFDMA, and beamforming support ● Omada SDN platform for cloud management and network control ● PoE support for flexible installation in areas with limited wiring ● Compact ultra-slim design for easy deployment in various environments	518.00	8	4144.00
Server 	Brand: Dell Model: PowerEdge R740 Features: ● High-performance PERC 12 RAID controllers ● Scalable design for mission-critical workloads and high-performance computing ● Balanced memory configuration ● Advanced reliability and robust quality control	19830.48	4	79321.92

	Customizable interleaving			
Firewall 	Brand: Fortinet Model: FortiGate 6000F Series Features: - Real-time anomaly detection and zero-trust access - AI-driven protection with FortiOS - Ultra-low latency and high-speed performance - Scalable security fabric for future-proofing enterprise networks - Comprehensive intrusion prevention system (IPS) for network safety	42402.18	4	169608.72
Personal Computer (PC) 	Brand: Lenovo Model: ThinkCentre M90s Gen 5 Features: 14th Gen Intel Core i5 processor - Supports up to 128GB DDR5 memory for multitasking efficiency - Integrated Intel UHD Graphics 770 - ThinkShield Security Suite - Compact design suitable for various office or lab setups	3358.00	190	638020.00
Monitor 	Brand: Lenovo Model: ThinkVision P27u-20 Features: - IPS (In-Plane Switching) for wide viewing angles (178° / 178°) - Response time in 4ms	2611.00	190	496090.00

	(Extreme Mode), 6ms (Normal Mode) ● Height-Adjustable ● Eye-Caring ● 3 years warranty ● Energy Star 8.0 (lower power consumption) ● Power consumption are 35W (typically), 240W (max)			
Projector 	Brand: BenQ MW612 Model: MW612 Features: ● Software-free wireless projection ● Stable video playback ● Dual HDMI Connectivity ● Automatically turns on projector once HDMI input signal detected ● Easy setup ● Kensington Lock for physical security to discourage theft ● Resolution: WXGA (1280x800) ● Light Source Life: Normal 4,000 hours, SmartEco 8,000 hours, LampSave 15,000 hours ● Picture Modes: 3D, Bright, Infographic, Presentation, sRGB, User 1, Vivid ● Brightness: 4,000 lumens ● Contrast Ratio: 20,000 : 1	2,584	7	18088.00
Smart TV 	Brand: Sony Model: Bravia XR X90K (85-inch) Features: ● Cognitive Processor XR™ for intelligent picture and sound optimization	16999	2	33,998.00

	● Google TV with access to a wide range of apps and voice control via Google Assistant ● Connectivity with Wi-Fi, ethernet, bluetooth, usb ports and HDMI inputs ● Display Technology: Full-Array LED with Local Dimming ● Resolution: 4K UHD (3840 x 2160) ● Audio Technology: Acoustic Multi-Audio, Dolby Atmos ● Additional Features: Voice search, Chromecast built-in, Apple AirPlay support, Energy-saving modes, Alexa Compatibility			
Smartboard 	Brand: Huawei Model: IdeaHub S2 (65-inch) Features: ● Display: 4K UHD resolution (3840x2160), multi-touch ● Integrated Software: Huawei CloudLink conferencing tools ● Connectivity: Wi-Fi 6, HDMI, USB, Bluetooth ● Audio/Video: Built-in 4K camera, 12-mic array ● Multi-touch support for interactive collaboration with others ● 8GB RAM and 64 GB Flash Memory	21,000	4	84,000.00
Microphone	Brand: Shure Model: MX418 Features: ● Type: 18-inch gooseneck condenser microphone	1,216	9	10,944.00

	● Polar Pattern: Cardioid (focused pickup from the front) ● Frequency Response: 50Hz to 17kHz ● Features: Lockable mute switch, LED indicator ● Audio Quality: High clarify for speech			
Connector 	Brand: Huawei Model: Cat6 UTP Connector Features: ● Compatibility: Support Cat6 cables, up to 1 Gbps ● Durability: High-quality plastic housing ● Installation: Tool-free design ● Standards: Meets TIA/EIA-568-C.2	3	400	1,200.00
Fiber Optic Cable 	Brand: Corning Model: ClearCurve OM4 Fiber Features: ● High Bandwidth Capacity: 10Gb/s over 550 meters, 1Gb/s over 1100 meters ● Superior bend performance, ideal for tight installation ● Core diameter: 50 microns ● Standards Compliance: ISO/IEC 11801, IEC 60793-2-10, TIA/EIA	4/meter	300	1,200.00
Total				1,609,792.22

3.3 Reflections

3.3.1 Are you surprised by the prices? How were you surprised?

Yes, the price of networking devices were surprising in several aspects. While I anticipated high prices for enterprise-grade equipment, a lot of the components including switches and firewalls were far beyond my expectations, in that they had highly sophisticated functionality and boasted the strongest performance metrics. For example, the Fortinet FortiGate firewall, priced at RM42,402.18 per unit, represents its advanced security features, which justify its high price but may appear steep at first glance. Some goods, like patch panels and connections, were less expensive than anticipated, demonstrating cost-effectiveness for their features and quality. The variability in costs highlights the importance of proper budgeting and cost-benefit analysis.

3.3.2 Have you ever considered cost as a factor for choosing networking devices?

Yes, pricing is always an important consideration when choosing networking devices, particularly for a project with a fixed budget like ours. Balancing performance and affordability is critical to meeting all objectives without exceeding the budget. For example, we chose the TP-Link Archer AX6000 router over the Asus RT-AX88U since it offered equal capabilities at a lesser price. Similarly, selecting bulk cables over pre-terminated ones enabled us to save money while maintaining flexibility and performance. Prioritizing critical features and analyzing alternatives enabled us to develop a cost-effective and scalable solution fit for the academic setting.

3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

Router:

For the router, we selected TP-Link Archer AX6000. Another router that can used to compared is Asus RT-AX88U router, and the details differences are at below:

Features	TP-Link Archer AX6000	Asus RT-AX88U
Type	Dual-Band Router	Dual-Band Router

Wifi Speed	WiFi 6 (2.4GHz) : up to 1148 Mbps WiFi 6 (5GHz) : up to 4804 Mbps	WiFi 6 (2.4GHz) : up to 1148 Mbps WiFi 6 (5GHz) : up to 4804 Mbps
Additional Ports	1 x USB 3.0	1 x USB 3.0, 1 x USB 2.0
Processor	Quad-core 1.7 GHz	Quad-core 1.8 GHz
MU-MIMO	Yes	Yes
OFDMA	Yes	Yes
Malware Protection	Trend Micro Protection	AiProtection Pro
Warranty	3-year limited warranty	2-year limited warranty
Price	RM1029	RM1620

Both TP-Link Archer AX6000 and the Asus RT-AX88U have similar functionality in terms of routers. Both of them used dual-band Wi-Fi 6 that provides up to 1148 Mbps for 2.4 GHz and up to 4804 Mbps for 6 GHz. Other than that, both provide MU-MIMO (Multi-User, Multiple Input, Multiple Output) that can ensure a router is able to communicate with multiple devices simultaneously by using multiple data streams. OFDMA (Orthogonal Frequency Division Multiple Access) is also being provided by both routers that help in breaking down data into small sub-channels and assigning them to different devices, and at the end, increase capacity and reduce latency in the faculty environment that will consist of many users at the same time. In terms of malware security, Trend Micro Protection in TP-Link Archer AX6000 provides basic security features such as antivirus, malware protection, and network security. However, AiProtection Pro in the Asus RT-AX88U provides the same security feature by Trend Micro but with additional features such as parental controls, deeper network monitoring, and intrusion detection. TP-Link Archer AX6000 provides a 3-year warranty, while Asus RT-AX88U provides a 2-year warranty. In terms of price, the TP-Link Archer AX6000 costs RM1029, and the Asus RT-AX88U costs RM1620.

Since the TP-Link Archer AX6000 provides similar Wi-Fi speed and security functionality with the Asus RT-AX88U, the **TP-Link Archer AX6000** is chosen due to its cost-effectiveness (less cost than the Asus RT-AX88U by 50%) and longer lifetime warranty.

Switch:

We choose the D-Link DGS-1250-52X as the switch chosen. The differences between the switch brands are stated below between Ubiquiti UniFi Switch Pro 48 500W and D-Link DGS-1250-52X:

Features	Ubiquiti UniFi Switch Pro 48 500W	D-Link DGS-1250-52X
Port Count	48 Ethernet ports, 2 SFP ports (1 Gbps), 2 SFP+ ports (1/10 Gbps)	48 Ethernet ports, 4 SFP+ ports (10 Gbps)
PoE Support	Yes	Yes
Switching Capacity	140 Gbps	176 Gbps
Packet Buffer Memory	16 Mbits	16 Mbits
Max Forwarding Rate	104.16 Mpps	130.95 Mpps

When choosing a switch, we need to consider the number of ports to make sure it can serve the building's needs. We consist of at least 30 PCs in a lab, thus we need to ensure there are more than 30 ports for the switch. Both the Ubiquiti UniFi Switch Pro 48 500W and the D-Link DGS-1250-52X consist of 52 ports, but the D-Link DGS-1250-52X has more SFP+ ports (10 Gbps) than the Ubiquiti UniFi Switch Pro 48 500W, which gives it more 10 Gbps performance. In terms of PoE support, both of them are able to provide simple PoE support, but the D-Link DGS-1250-52X provides less power support compared to Ubiquiti. It is acceptable because the switch is used for the purpose of education only. Additionally, both of them consist of 176 Gbps switching capacity and 16 Mbits packet buffer memory, which makes them able to handle large traffic loads in the lab environment. Lastly, the D-Link DGS-1250-52X consists of a maximum forwarding rate of 130.95 Mpps, which makes it able to handle more packets per second compared to the Ubiquiti UniFi Switch Pro 48 500W.

Thus, we choose the **D-Link DGS-1250-52X** as the switch to be used in this faculty environment.

Patch Panel:

We compare 24 Ports CAT6 RJ45 Keystone Patch Panel and NavePoint 24-Port CAT6 UTP Patch Panel:

Features	FS Cat6 24-Port Patch Panel	NavePoint 24-Port CAT6 UTP Patch Panel
Connector Types	RJ45 Ethernet/HDMI/USB	RJ45
Number of Ports	24	24
Mounting Type	Rack or Cabinet	Rack Moounted
Cable Types	Cat6/Cat5e	Cat 6
Shielding Type	C6P-U24FT1U: Unshielded C6P-S24FT1U: Shielded	UTP
Dimension	1.75"x19"x3.62" (44.5x482.5x92mm)	1.75"x19"x1.97" (44.5x482.5x50mm)
Price	RM312.00	RM209.58

In terms of patch panels, the FS Cat6 24-Port Patch Panel supports RJ45 (Ethernet), HDMI, and USB (via adapters) connector types, while the NavePoint 24-Port Cat6 UTP Patch Panel only provides RJ45 (Ethernet). Thus, the FS Cat6 24-Port Patch Panel provides more flexibility in cabling by providing HDMI and USB connector types. Both of them consist of 24 ports and can be rack compatible. In terms of cable types, the NavePoint 24-Port Cat6 UTP Patch Panel only supports Cat6 cables, but the FS Cat6 24-Port Patch Panel provides both Cat6 and Cat5e cables. In terms of shielding type, one version of the FS Cat6 24-Port Patch Panel (C6P-S24FT1U) provides shielding, which offers protection against electromagnetic interference, while the other version and the NavePoint 24-Port Cat6 UTP Patch Panel are unshielded. Additionally, the FS Cat6 24-Port Patch Panel is larger and more costly than the NavePoint 24-Port CAT6 UTP Patch Panel. In short, although FS Cat6

24-Port Patch Panels are more expensive, their flexibility and features make them more suitable to be used in a university environment; thus, **FS Cat6 24-Port Patch Panels** are chosen.

Cable:

We selected Cable Matters Riser Rated (CMR) Cat6 Bulk Ethernet Cable as the cable used in the lab environment. The differences between this cable and the Monoprice Cat6 Component Level Ethernet Cable are listed below:

Features	Cable Matters Riser Rated (CMR) Cat6 Bulk Ethernet Cable	Monoprice Cat6 Component Level Ethernet Cable
Type	Bulk Cable	Pre-terminated patch cable
Length	1000 feet (305 meter)	From 0.5ft to 50ft
Speed	Up to 10Gbps	Up to 1GBps
Bandwidth	Up to 550MHz	250MHz (up to 550MHz)
Installation	Need cutting, crimping and termination	Plug-and-play, no additional tools required
Price	RM1888.73 for 1000ft (RM1.89/ft)	RM22.44 for 5ft (RM4.49/ft)

There are two types of cable: one is bulk, and the other one is pre-terminated. For bulk cable, such as Cable Matters Riser Rated (CMR) Cat6 Bulk Ethernet Cable, it will be required to cut, crimp, and terminate on our own, but pre-terminated cable, such as Monoprice Cat6 Component Level Ethernet Cable, can let us easily plug in. In between these two types of cable, bulk cable is more suitable since it allows us to cut the exact lengths of cable to connect with those PCs and network devices in the lab to the patch panel and minimise the waste compared to pre-terminated cable. In terms of performance, Cable Matters Riser Rated (CMR) Cat6 Bulk Ethernet Cable provides a higher performance where its speed is up to 10Gbps and bandwidth up to 550MHz, but Monoprice Cat6 Component Level Ethernet Cable only provides 1Gbps speed and the same maximum bandwidth. Additionally, although the total price of Cable Matters Riser Rated (CMR) Cat6 Bulk Ethernet Cable is expensive, it

provides a lower price per foot compared to Monoprice Cat6 Component Level Ethernet Cable. In a nutshell, Cable Matters Riser Rated (CMR) Cat6 Bulk Ethernet Cable is more suitable to use in the lab environment compared to Monoprice Cat6 Component Level Ethernet Cable that is more suitable to use in an individual environment.

Modem:

We selected ARRIS SURFboard SB8200 DOCSIS 3.1. The difference between ARRIS SURFboard SB8200 DOCSIS 3.1 and other different brands we can compare is Motorola MB7621 Cable Modem.

Features	Motorola MB7621 Cable Modem	ARRIS SURFboard SB8200 DOCSIS 3.1
DOCSIS Standard	DOCSIS 3.0	DOCSIS 3.1
Max Speed	900 Mbps	2 Gbps
Ethernet Ports	1x Gigabit Ethernet	2x Gigabit Ethernet
Compatibility	Comcast Xfinity, Cox, Spectrum, and more	Major U.S. cable providers (Xfinity, Spectrum, etc.)
Target Users	Suitable for smaller teams or low-to-moderate usage environments	Ideal for high-traffic environments, such as faculty computing labs, research groups, and high-bandwidth applications
Reliability for Multiple Devices	Suitable for up to moderate usage, good for home office settings	Better suited for handling multiple devices, ideal for larger teams and environments with higher concurrent usage
Additional Features	Ideal for fast internet plans, single Ethernet port	Supports high-demand applications like VR gaming

		and 4K streaming, dual Ethernet ports for increased connectivity flexibility
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The ARRIS SURFboard SB8200 is perfect for faculty computing environments with speeds up to 2 Gbps to support huge file transfers, video conferencing, and a multitude of devices. The dual Ethernet ports allow running separate networks, such as one network for faculty and another network for students. In addition, this modem supports DOCSIS 3.1, making it future-proof. It's perfect for high-demand settings with many users, like labs and lecture halls. By contrast, the Motorola MB7621 will serve for smaller setups with lower speeds, such as general browsing and email. Overall, the **ARRIS SURFboard SB8200** is a better option to ensure high-speed, reliable internet in busy environments.

Access Point:

For the access point used in this project, we use TP-Link EAP610 Ultra-Slim. The major difference between TP-Link EAP610 Ultra-Slim and other different brands we can compare is Ubiquiti UAP-AC-PRO.

Features	TP-Link EAP610 Ultra-Slim	Ubiquiti UAP-AC-PRO
Speed	Up to 1800 Mbps	Up to 1300 Mbps
Wi-Fi Technology	Wi-Fi 6 (MU-MIMO, OFDMA), Beamforming, Mesh	MU-MIMO, 802.11ac 3x3 MIMO
Frequency Band	2x Gigabit Ethernet	Dual-Band (2.4 GHz & 5 GHz)
Security	Ideal for high-traffic environments, such as faculty computing labs, research groups, and high-bandwidth applications	Suitable for smaller teams or low-to-moderate usage environments
Cloud Management	Omada SDN Platform, Cloud Management	UniFi Controller, Cloud Management
Installation Ease	Easy (Managed via Omada app, intuitive setup)	Moderate (Requires UniFi Controller)
Additional Features	Supports high-demand applications like VR gaming and	Ideal for fast internet plans, single Ethernet port

	4K streaming, dual Ethernet ports for increased connectivity flexibility	
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Faculty computing would be fit for the **TP-Link EAP610 Ultra Slim** because Wi-Fi 6 support can give better efficiency and high performance for devices connected in that particular place. With 1800 Mbps, it is designed for high-bandwidth activities including online lectures and research. Apart from cloud management available on the Omada SDN, platform remote networks control is given easily to flexible PoE support installment given in such areas where the wiring facility is not available enough. These features make it a future-proof solution, providing stable and efficient connectivity for a growing network.

Server:

For the server used in this project, we have chosen Dell PowerEdge R740 . The major difference between with other different brands we can compare is HPE ProLiant DL380 Gen10.

Features	Dell PowerEdge R740	HPE ProLiant DL380 Gen10
Performance	20% better performance in benchmarks, lower latency with PERC 12 RAID controllers	Customizable interleaving for power efficiency and specific workloads
Memory Configuration	Focus on balanced configurations to maximize bandwidth	Reliable, but more reported downtime compared to Dell
Reliability & Uptime	Fewer issues, better real-world uptime, robust quality control	Dual-Band (2.4 GHz & 5 GHz)
RAID Controllers	PERC 12 (Broadcom SAS4116W) offers better throughput and lower latency	SmartArray RAID controllers, good performance but not as fast as Dell's
Management Software	Dell OpenManage, functional but less intuitive	HPE iLO, known for its intuitive and quick navigation
Scalability	Good, focused on performance and reliability	Highly scalable, ideal for hybrid IT and cloud solutions
Additional Features	Mission-critical workloads, high-performance computing, reliable uptime	Flexible, scalable environments needing adaptability and cloud integration

Dell PowerEdge is still the best in reliability and performance, especially for research or high-demanding activities when it comes to faculty computing. It ensures that the operations run smoothly with very minimal downtime. HPE ProLiant will be a good option if one needs more scalability and flexibility for cloud or hybrid setups, but it may result in slightly more downtime and higher latency. Go with Dell PowerEdge for reliability or HPE ProLiant for flexibility and scalability.

Firewall:

For the firewall used in this project, we have chosen Fortinet FortiGate 6000F Series. The major difference between with other different brands we can compare is Netgate pfSense Security Gateway Appliances.

Features	Fortinet FortiGate 6000F Series	Netgate pfSense Security Gateway Appliances
Target Environment	Large enterprises, service providers, high-demand computing.	Small to mid-sized enterprises, remote offices, and branch setups.
Performance	Ultra-low latency, synergy processing unit for high-speed performance.	Powered by Intel processors (Cortex, Atom, Xeon), with up to 16GB RAM.
Security Features	Real-time anomaly checks, intrusion prevention, zero-trust access.	Real-time alerts, security monitoring with open-source pfSense software.
Management	FortiOS and AI-driven protection. Fortinet security fabric.	Open-source pfSense with commercial support, flexible customization.
Cost-Performance	High initial investment, but extremely efficient for large, complex environments. The cost is justified by the enterprise-grade security, performance, and scalability.	Excellent cost-performance ratio, especially for small to mid-sized setups. Offers open-source flexibility, lower upfront cost, and good support for general use cases.

The **FortiGate 6000F Series** is suited for large educational institutions with very high security requirements, powered with high performance, low latency, and all security features. It will be the best fit for environments that need robust security, scalability, and future-proofing.

Personal Computer (PC):

For the Personal Computer (PC) used in this project, we have chosen Lenovo ThinkCentre M90s Gen 5. The major difference between Lenovo ThinkCentre M90s Gen 5 with other different brands we can compare is Dell OptiPlex 5090 Tower.

Features	Lenovo ThinkCentre M90s Gen 5	Dell OptiPlex 5090 Tower
Processor	14th Generation Intel® Core™ i5-14400 Processor	11 th Generation Intel® Core™ i5 14500 vPro®
Memory	Up to 128GB DDR45 4400MHz	Up to 64GB DDR4 3200MHz
Graphics	Integrated Intel UHD Graphics 770	Integrated Intel UHD Graphics 750
Security	TPM 2.0, ThinkShield Security Suite	TPM 2.0, Dell SafeBIOS
Price	RM3358	RM 3929

The Lenovo ThinkCentre M90s Gen 5 offers advanced hardware than the Dell OptiPlex 5090 Tower. This means as much as up to 14th Generation Intel Core i5 processors for better performance compared to Dell's offering of up to 11th Generation. In addition, Lenovo supports up to 128GB of DDR5 memory, which is faster and higher capacity than Dell's 64GB DDR4 maximum. In terms of graphics, Lenovo's integrated Intel UHD Graphics 770 outclasses Dell's Intel UHD Graphics 750. For security, both have TPM 2.0, but Lenovo takes a further step with its ThinkShield Security Suite. Pricing for Lenovo is affordable and lower compared to Dell. Hence, the **Lenovo ThinkCentre M90s Gen 5** is a more capable choice because of its greater processing performance, higher memory capacity, and enhanced security features.

Monitor:

For the monitor used in this project, we have chosen Lenovo ThinkVision P27u-20. The major difference between Lenovo ThinkVision P27u-20 with other different brands we can compare is Dell UltraSharp U2723QE.

Feature	Dell UltraSharp U2723QE	Lenovo ThinkVision P27u-20
Display Type	IPS Black Technology	IPS Technology
Response Time	8ms GtG	4ms (Extreme Mode), 6ms (Normal Mode)
Connectivity	HDMI (HDCP 2.2), DisplayPort 1.4 (HDCP 2.2), DisplayPort output, USB-C upstream/DisplayPort 1.4 Alt Mode with Power Delivery (power up to 90W, HDCP 2.2), USB-C 3.2 Gen 2 upstream, USB-C 3.2 Gen 2 downstream (power up to 15W), 4 x USB 3.2 Gen 2 downstream, USB 3.2 Gen 2 downstream with Battery Charging 1.2, Audio line-out (mini-jack), LAN (RJ-45)	1 x Thunderbolt 4 in, 1 x Thunderbolt 4 out, 2 x HDMI2.0, 1 x DP1.2, 1 x RJ45, 3 x USB 3.2 Gen1 with 1 x BC 1.2 (Downstream), 1 x USB 3.2 Gen1 Type-C (data & power delivery up to 27W, downstream) 1 x USB 3.2 Gen1 Type-B (Upstream) 1 x Audio Out (3.5 mm)
Price	RM 3039	RM 2611

Based on the comparison table, The Dell UltraSharp U2723QE offers advanced IPS Black Technology, which provides deeper blacks and improved contrast for high-quality visuals. However, its response time is slower at 8ms GtG, making it less suitable for fast-paced tasks or gaming compared to the Lenovo ThinkVision P27u-20, which has 4ms in Extreme Mode. In terms of connectivity, Dell provides more USB-C ports and robust power delivery options (up to 90W), making it better suited for modern setups requiring multiple devices and power over USB. Meanwhile, Lenovo includes Thunderbolt 4 ports, which are advantageous for high-speed data transfer and daisy-chaining multiple monitors. Besides, Lenovo is

significantly more affordable compared to Dell's RM 3039, making it a better value offer. Therefore, the **Lenovo ThinkVision P27u-20** is a better choice for a cost-effective, high-performance monitor with excellent connectivity and faster response time.

Projector:

For the projector used in this project, we have chosen BenQ MW612. The major difference between BenQ MW612 with other different brands we can compare is Epson EB-X51.

Feature	Epson EB-X51	BenQ MW612
Display Technology	3LCD Technology	DLP Technology
Resolution	XGA (1024 x 768)	WXGA (1280 x 800)
Brightness	3,800 lumens	4,000 lumens
Contrast Ratio	16,000:1	20,000:1

The Epson EB-X51 employs 3LCD technology, giving it brighter and truer colours and making it great for rich visual presentations. The BenQ MW612, meanwhile, employs DLP technology, ensuring creepier images with better contrast. This is better suited for text-oriented presentations and video playback. Next, the native resolution of BenQ MW612 is WXGA (1280 x 800), which is quite higher than the Epson EB-X51 with XGA (1024 x 768) that allows for more clear images. Besides, BenQ is brighter with 4,000 lumens, and the Epson is at 3,800 lumens, while it also has better contrast (20,000:1 vs. 16,000:1), giving it richer black levels and better depth of images. Hence, the **BenQ MW612** is the preferred choice for its superior specifications.

Smart TV:

For the projector used in this project, we have chosen Sony Bravia XR X90K. The major difference between Sony Bravia XR X90K with other different brands we can compare is LG OLED C2.

Feature	Sony Bravia XR X90K	LG OLED C2
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Display Technology	Full-Array LED with Local Dimming	OLED evo
Resolution	4K UHD (3840 x 2160)	4K UHD (3840 x 2160)
Smart Features	Google TV, Chromecast Built-in, Alexa Compatibility	webOS, ThinQ AI, Alexa Compatibility
Audio Technology	Acoustic Multi-Audio, Dolby Atmos	AI Sound Pro, Dolby Atmos

The Sony Bravia XR X90K features Full-Array LED with Local Dimming that produces decent bright performance and uniform contrast, which makes it a great option for classroom and labs environments, the LG OLED C2's OLED evo technology allows for true blacks with amazing contrast, especially in darker environments. Both the TVs provide support for has 4K UHD resolution (3840 x 2160) to produce crisp, detailed images suitable for presentations and multimedia use. Concerning smart features, the Sony Bravia XR X90K runs on Google TV with Chromecast Built-in, this makes it easily integrated with devices and applications applicable in teaching and collaboration environments. On the other hand, the LG OLED C2 operates on webOS with ThinQ AI that supports making the ecosystem well-integrated with smart home products, besides giving leadership experience. From the two, they both support Dolby Atmos to provide immersive audio, but the Sony one features Acoustic Multi-Audio for sound that is in sync with the visuals to enhance large rooms experience. Therefore, the **Sony Bravia XR X90K** is the optimal choice due to its balanced performance, practical features, and cost-effectiveness, making it well-suited for enhancing learning and collaboration.

Smartboard:

For the smartboard we used in this project, we have chosen Huawei IdeaHub S2. The major difference between Huawei IdeaHub S2 with other different brands we can compare is Cisco Webex Board 55S.

Feature	Huawei IdeaHub S2	Cisco Webex Board 55S
Display	4K UHD, multi-touch, 65-inch	4K UHD, multi-touch, 55-inch

Integrated Software	Huawei CloudLink conferencing tools	Webex Teams integration
Connectivity	Wi-Fi 6, HDMI, USB, Bluetooth	Wi-Fi 6, HDMI, USB, Ethernet
Audio/Video	Built-in 4K camera, 12-mic array	Built-in 4K camera, integrated mic
Price (MYR)	21000	26000

Both the Huawei IdeaHub S2 and Cisco Webex Board 55S are technologically rich smartboards, each having a 4K UHD display with multitouch and advanced collaboration capabilities. The Huawei IdeaHub S2 will be more suitable for bigger classrooms because it has a slightly larger display, 65-inch versus 55-inch. It comes fitted with CloudLink conferencing tools and thus fits best in hybrid classrooms. It also features Wi-Fi 6, HDMI, USB, and Bluetooth for flexible connectivity, so much needed in dynamic learning environments. The powerful nature of the Cisco Webex Board 55S makes it best fit for corporate environments that have previously established a Cisco infrastructure. The small size of the display further limits its appropriateness in large-classroom settings. This will make finding a balance between cost and features, thus setting the **Huawei IdeaHub S2** as better suited for classrooms.

Microphone:

For the microphone we used in this project, we have chosen Shure MX418. The major difference between Shure MX418 with other different brands we can compare is Audio-Technica ATUC-M43H.

Feature	Shure MX418	Audio-Technica ATUC-M43H
Type	18-inch gooseneck condenser microphone	17-inch gooseneck condenser microphone
Polar Pattern	Cardioid (focused pickup from the front)	Hypercardioid (narrower pickup pattern)
Frequency Response	50Hz to 17 kHz	30Hz to 20kHz
Features	Lockable mute switch, LED indicator	Modular design, detachable microphone unit

Audio Quality	High clarity for speech	Wide frequency range
Price (MYR)	1216	1500

The cardioid pattern of the Shure MX418 is great for general-purpose speech applications, providing a good balance of focus and ambient noise rejection. The Audio-Technica ATUC-M43H has a hypercardioid pattern, which provides a narrower field of pickup and is more effective in environments with significant noise or multiple microphones in use. Then, Shure MX418 is tuned specifically for speech clarity, 50 Hz to 17 kHz, making it perfect for educational and conferencing purposes. For usability, Shure MX418 has a lockable mute switch and LED indicator-user-friendly features at live presentations-while the Audio-Technica ATUC-M43H has a modular design for flexibility in maintenance and system upgrades. Therefore, **Shure MX418** is chosen as we will use this device for simple conference setups, and it is cost-effective.

Connector:

For the connector we used in this project, we have chosen Huawei Cat6 UTP Connector. The major difference between Huawei Cat6 UTP Connector with other different brands we can compare is Cisco RJ45 Modular Plug.

Feature	Huawei Cat6 UTP Connector	Cisco RJ45 Connector
Compatibility	Support Cat6 cables, up to 1 Gbps	Support Cat6 cables, up to 10 Gbps
Durability	High-quality plastic housing	Gold-plated contacts for durability
Installation	Tool-free design	Requires standard crimping tools
Standards	Meets TIA/EIA-568-C.2	Exceed TIA/EIA-568-C.2
Price (MYR)	3	5

The connectors are important for ensuring stable network connections. Huawei Cat6 UTP Connectors provide adequate performance for most educational environments where speeds up to 1 Gbps are sufficient and are easy to install with a tool-free design. Cisco RJ4 Modular Plug supports higher speeds than is actually required for a university and thus will result in a

waste of resources. This comes with gold plating to make it resistant to corrosion. This can be less convenient in a lab environment since the installation requires crimping tools. For the sake of the requirement for moderate performance, it would be advisable to go for **Huawei Cat6 UTP Connectors**.

Fiber Optic Cable:

For the fiber optic cable we used in this project, we have chosen Corning ClearCurve OM4 Fiber. The major difference between Corning ClearCurve OM4 Fiber with other different brands we can compare is Prysmian Draka Multimode Fiber.

Feature	Corning ClearCurve OM4 Fiber	Prysmian Draka Multimode Fiber
Type	Multimode	SingleMode
Bandwidth capacity	Up to 550 meters at 10Gbps	Up to 55 meters at 10Gbps
Core diameter	50 microns	50 microns
Bend radius	Ultra-low bend radius for tight installations	Low bend radius, optimized for flexibility
Environment suitability	Ideal for tight indoor setups	Excellent for structured cabling in larger building

The **Corning ClearCurve OM4 Fiber** has the best performance and flexibility for installation in classrooms, labs, and video conferencing rooms. These cables have an ultra-low bend radius, which allows them to be installed around dense and complex lab spaces without much problem. This way, cables can go around conduits, corners, and tight cable trays without being assessed any performance penalties. This cable will go through data transmissions to the tune of up to 10 Gbps over 550 meters and certainly allows for enough bandwidth to support high-speed internet, real-time video conferencing, and other demanding applications like data analysis in labs.

Meeting Minutes #1

Date/Time		31 Dec 2024 12:00pm	
Location		Online (Google Meet)	
Agenda		1. Discuss the question on Task 3, 4 and 5	
		2. Discuss report format	
		3. Task Separation and Distribution	
Meeting Host		Goh Jiale	
Attendance			
Name		Time	Reason For Absence
Goh Jiale		12:00pm	-
Joanne Ching Yin Xuan		12:00pm	-
Evelyn Goh Yuan Qi		12:00pm	-
Lim Yu Han		12:00pm	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Discuss of question on Task 3, 4 and 5	Goh Jiale briefly explains Case Study and Task 1 deliverables.	Goh Jiale
		All members research and list down a list of devices that need to be included in the building.	All Member
		Evelyn suggests that the floor diagram can use the original diagram in task 1 that draws in Cisco Packet Tracer.	Evelyn
3	Discuss Report Format	Joanne and Lim Yu Han list down the task 3, 4 and 5 deliverables	All Member
		All members decide to compare the network devices in the questions section instead of introduction.	
5	Task Distribution	Compare Smartboard, Microphone,	Yuhan

		connectors and Fiber Optic Cable + Table of Content	
		Compare Modem, Access Point, Server and Firewall + Introduction	Joanne
		Compare PC, Monitor, Projector and Smart TV + Intro + Reflection question 1 and 2.	Evelyn
		Compare Router, Switch, Patch Panel and Cable + Table of Content	Goh Jiale
8	Meeting ended	Lim Yu Han will do the next meeting minute.	Lim Yu Han
		The next meeting is on 4th November at 5pm through Google Meet (online).	All Member

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